

Carlos Aguilera

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EDUCATION

California State University, Long Beach

Class of 2026

Bachelor of Science in Computer Science - 3.52 GPA

Long Beach, CA

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Databases, Object-Oriented Programming, Computer Architecture & Organization, Computer Networks, Discrete Math

EXPERIENCE

Software Engineer

May 2022 – Present

Ethika Inc.

Lake Forest, CA

- Built a **LangChain-powered AI assistant** for internal teams, giving it tools to understand database structure, relationships, and business context so employees could pull reports, insights, or analyze data.
- Executed a three-stage migration from Imgix to a **CloudFront + S3 architecture**, moving 150k+ assets with zero downtime. **Saving 36k a year.**
- Led the migration from Algolia to TypeSense, creating a bridge between a legacy tag-based system and Algolia. I developed a **query optimization** approach that streamlined three calls into one, **saving \$75k annually.**
- Developed a Music CMS system with a **RabbitMQ** task queue for performing **asynchronous stem extraction**. I eliminated an inefficient process involving multiple Dropbox accounts for more than **10k artists.**
- Engineered an all-in-one return system that incorporates **3PL webhooks** and simple labeling. Created an idempotent system for refunds that synchronizes state between the portal and backend systems, **automating a high-volume manual workflow.**

Software Engineer

June 2025 – Present

The Inked Network

Remote

- Built an **AI-driven media pipeline** processing **130k images** monthly, integrating OpenAI tagging, ImageKit/S3 delivery, and AWS Rekognition for automated moderation.
- Migrated a legacy database to a redesigned schema using custom backfill and cleanup scripts, enabling **secure row-level access** controls across all tables.
- Architected end-to-end **Stripe subscription infrastructure**, implementing webhooks, subscription state handling, and Apple-compliant external payment flows.
- Designed a **real-time chat system** using Supabase Realtime with secure RLS policies, delivery/read receipts, and optimized channel filtering.
- Established CI/CD with preview environments, **automated database seeding**, Xcode Cloud, running tests, and rollback strategies.

PROJECTS

Software Developer

July 2023 – July 2024

Robotics Club

Saddleback College

- Designed a **high-availability Base Station GUI** using Django and ROS2, integrating real-time GPS telemetry and heading data to facilitate autonomous rover traversal.
- Engineered a **low-latency telemetry dashboard** with live camera feeds, directly contributing to the team's qualification for the URC Finals and a **World Rank of 17th** (up from 20th).
- Implemented **asynchronous data processing** to handle high-frequency sensor streams, ensuring precise navigation and operator feedback during competitive runs.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, PHP, Java, C++, SQL, HTML, CSS

Frameworks & Libraries: Vue.js, Nuxt, React, Tailwind CSS, Symfony, Django, FastAPI, NestJS, Node.js, GraphQL, Capacitor

Databases & Authentication: PostgreSQL, MongoDB, Redis, Supabase, JWT, OAuth

Cloud & DevOps: AWS, Google Cloud, Azure, Cloudflare, Vercel, Docker, Linux, Nginx, Git, CI/CD, CloudWatch

Platforms & Integrations: Stripe, App Store Connect, Google Play Console, REST APIs, Web Services

Engineering Concepts: Microservices, Event-Driven Systems, Object-Oriented Programming, Data Modeling, Testing, Debugging, Agile